

MODEL QUESTION PAPER

TED (15)5071

FIFTH SEMESTER DIPLOMA EXAMINATION IN CHEMICAL ENGINEERING

MASS TRANSFER OPERATIONS-1

(Graph sheet should be issued)

Time: 3hrs

Maximum marks: 100

PART –A

Maximum marks: 10

I Answer the following questions in one or two sentences. Each question carries 2 marks

- 1 State Fick's law.
- 2 Define humidity.
- 3 Suggest suitable dryer for (1) milk powder (2) wet lumpy solids
- 4 State Raoult's law.
- 5 Define more volatile component. (5X2=10)

PART –B

(Maximum marks:30)

II Answer any five of the following questions. Each question carries 6 marks.

- 1 Explain the mechanism of absorption operation. 6
- 2 Differentiate between physical and chemical adsorption. 6
- 3 Summarise the working of counter flow induced draft cooling tower. 6
- 4 Explain the working of tray drier 6
- 5 Differentiate between bound water and unbound water 6
- 6 Derive Rayleigh's equation. 6
- 7 Illustrate the equilibrium distillation. 6

(5X6=30)

PART – C

Maximum marks:60

(Answer one full question from each unit. Each full question carries 15 marks)

Module –I

- III (a) Derive the material balance equation relating to absorption. 7
 (b) What are the factors to be considered for the selection of solvent in gas absorption? 8

OR

- IV (a) Explain the constructional details and working of packed column. 8
 (b) Ammonia from ammonia air mixture is to be absorbed in an absorption tower using water as solvent. Data for absorption system is as follows 7
 Air flow rate = 500 Kg/hr
 Liquid phase compositions:
 At the top of packing = 0.000019 Kg NH₃/Kg H₂O
 At the bottom of packing = 0.0008 Kg NH₃/Kg H₂O
 Gas phase compositions:
 At the bottom of the packing = 0.0094 Kg NH₃/Kg inert gas
 At the top of packing = 0.00077 Kg NH₃/Kg inert gas
 Compute the flow rate of water entering the absorption tower.

Module II

- V (a) Explain the constructional details and working of Atmospheric type cooling towers. 8
 (b) A sample of hydrogen at 20 ° C and 101.325 KPa pressure contains CS₂ vapour, such that its dew point is 0 ° C. Calculate (1) Molal humidity (2) Absolute humidity (3) Relative humidity and (4) Percentage humidity 7
 Data given
 Vapour pressure of CS₂ at 0 ° C = 16.95 KN/m²
 Vapour pressure of CS₂ at 20 ° C = 23.33 KN/m²

OR

- VI (a) Explain constructional details and working of humidifiers 8
 (b) Summarise the factors to be considered for the selection of cooling towers 7

Module -III

- VII (a) Explain the constructional details and working of spray dryer 6

- (b) A batch dryer removes water from a solid at the rate of 15Kg/hr during constant rate period. The critical moisture content on dry basis is 0.5 and equilibrium moisture content is 0.04 on dry basis. 75Kg of dry solid with 50Kg water is entering the dryer. How long will be the drying take place if final product contains 0.08Kg of water/Kg of dry solid
OR
- VIII (a) Explain the constructional details and working of Rotary dryer. 5
- (b) A wet solid is dried from 30% to 10% under constant drying condition in 6 hrs. The equilibrium moisture content is 2% and critical moisture content is 16%. How long will it take to dry the material from 15% to 6% moisture content under the same drying condition. All moisture contents on dry basis 10

Module –IV

- IX (a) Derive an equation for relative volatility from volatility of A and B. 6
- (b) Liquid mixture containing 30mole% methanol and 70mole% water is fed to differential distillation at atmospheric pressure with 60mole% of the liquid is distilled. Find the composition of the composed distillate and residue. Data given 9

x	0	0.05	0.1	0.2	0.4	0.6	0.8	0.9	1
Y	0	0.13	0.21	0.37	0.6	0.77	0.9	0.95	1

OR

- X (a) The vapour pressure of n-heptane (A) and n-octane (B) are given in the table. Assume that Raoult's law and Dalton's law can apply. Compute vapour –liquid equilibria at constant pressure of 101.325KPa and plot the boiling point diagram and equilibrium diagram. 9

Temperature °C	Vapour pressure of n-heptane KPa	Vapour pressure of n-octane KPa
98.4	101.325	44.396
105	125.323	55.595
110	139.988	64.528
115	159.987	74.795
120	179.985	86.659

125.6	205.316	101.325
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(b) Summarise the industrial applications of steam distillation .

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